



Chapter 12: Physical Storage Systems

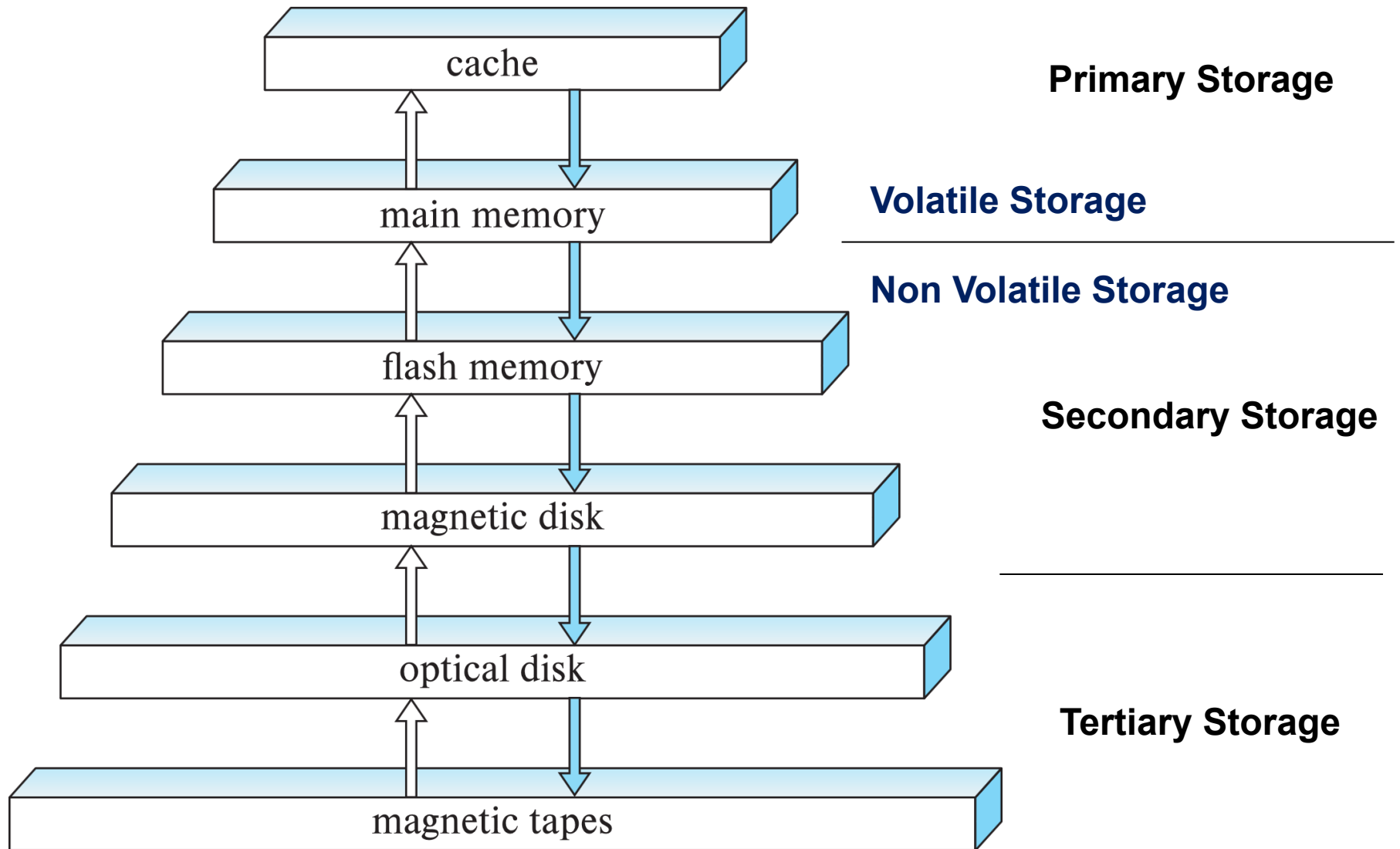
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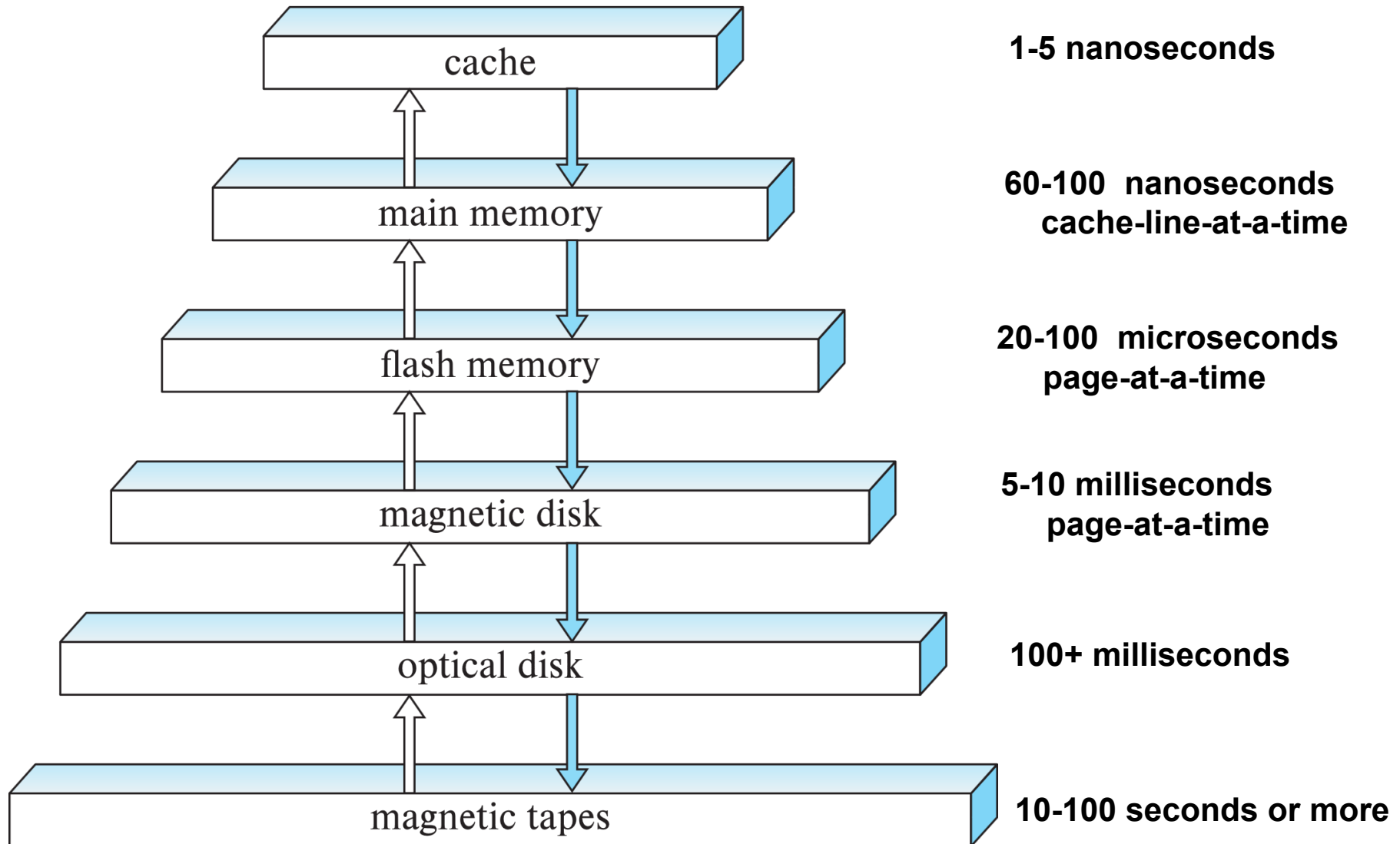


Storage Hierarchy





Storage Hierarchy: Access Time



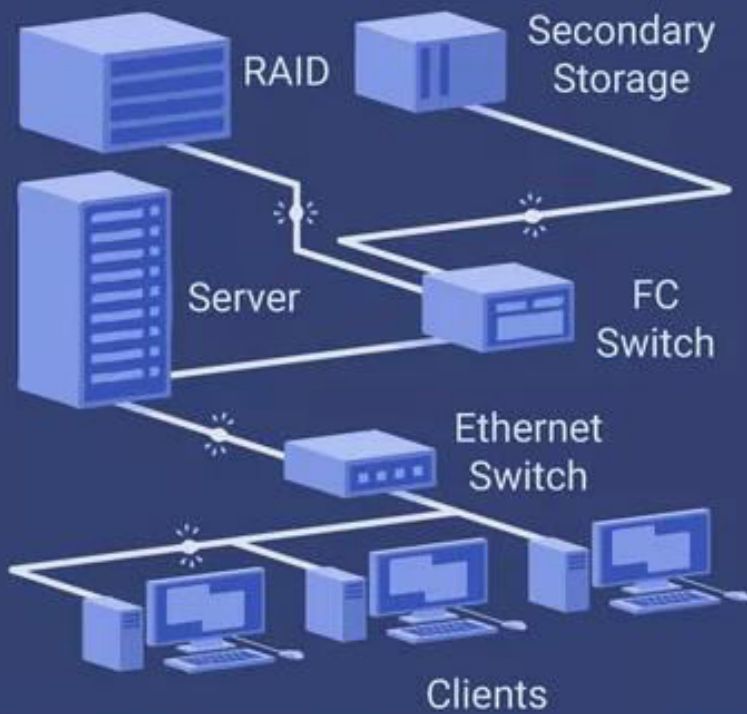


Storage Interfaces

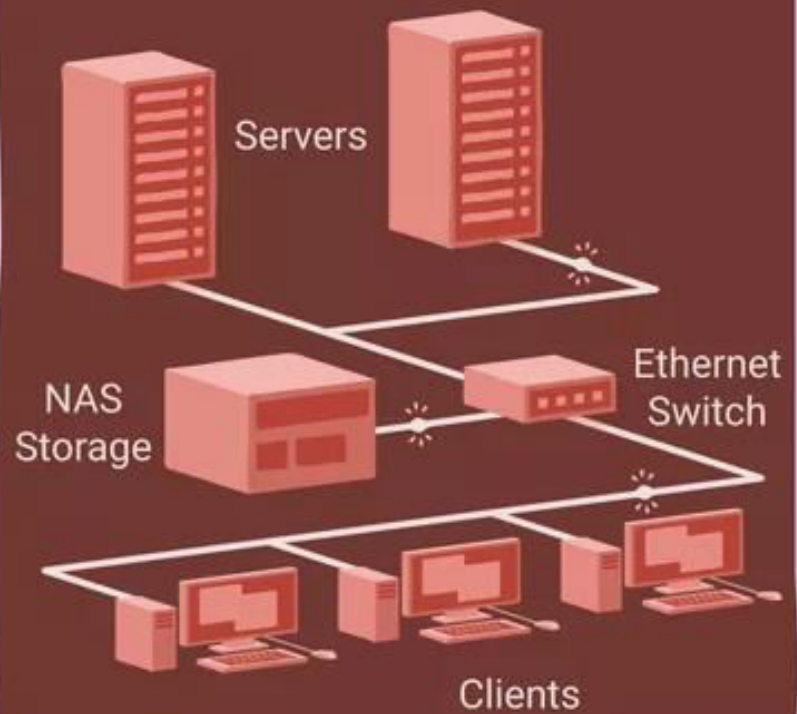
- Disk interface standards families
 - **SATA** (Serial ATA)
 - SATA 3 supports data transfer speeds of up to 6 gigabits/sec
 - **SAS** (Serial Attached SCSI)
 - SAS Version 3 supports 12 gigabits/sec
 - NVMe (Non-Volatile Memory Express) interface
 - Works with PCIe connectors to support lower latency and higher transfer rates
 - Supports data transfer rates of up to 24 gigabits/sec
- Disks usually connected directly to computer system
- In **Storage Area Networks (SAN)**, a large number of disks are connected by a high-speed network to a number of servers
- In **Network Attached Storage (NAS)** networked storage provides a file system interface using networked file system protocol, instead of providing a disk system interface



SAN

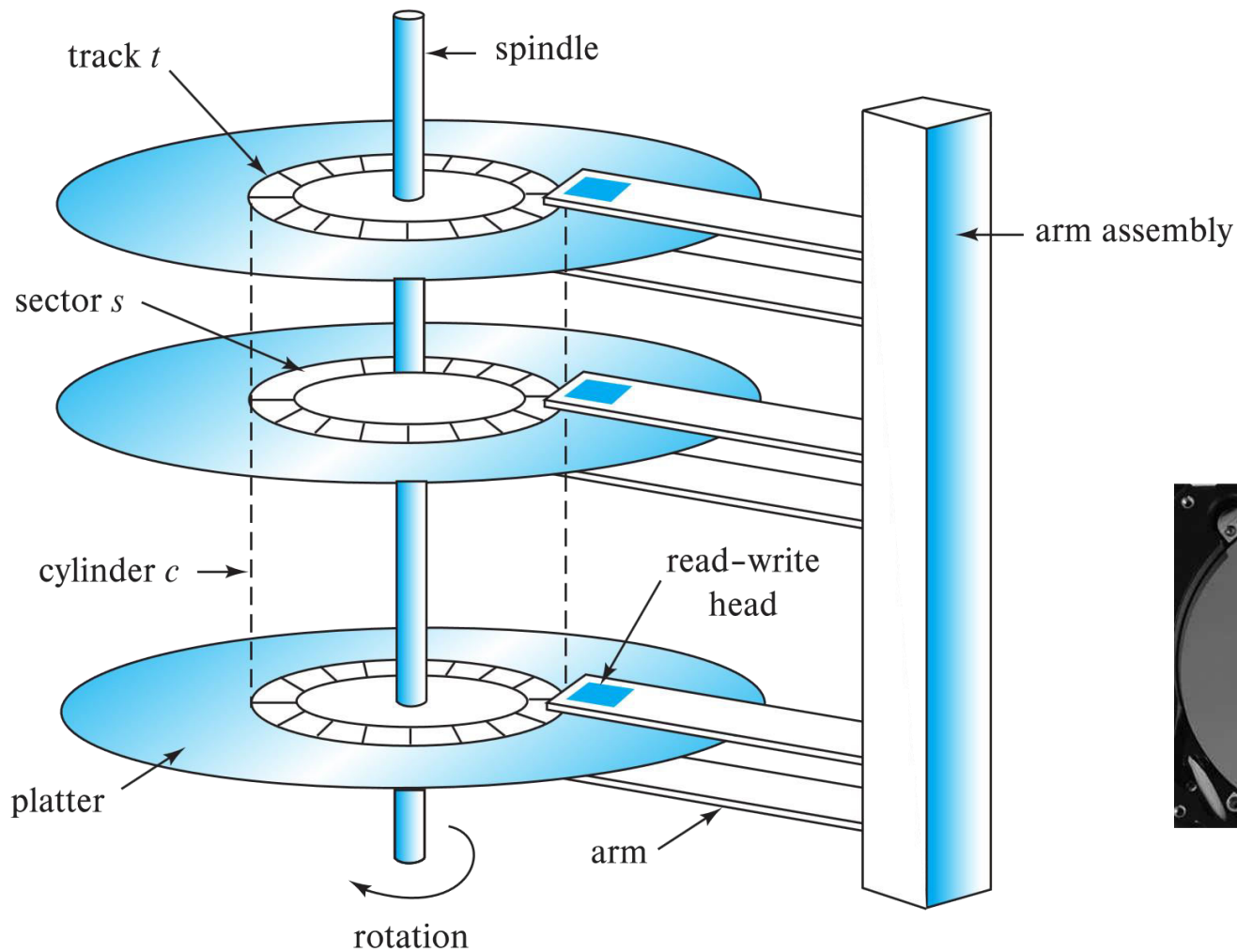


NAS





Magnetic Hard Disk Mechanism



Schematic diagram of magnetic disk drive

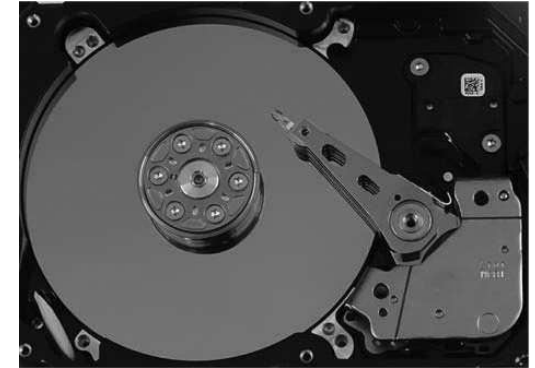


Photo of magnetic disk drive



Magnetic Disks (Cont.)

- **Disk controller** – interfaces between the computer system and the disk drive hardware.
 - accepts high-level commands to read or write a sector
 - initiates actions such as moving the disk arm to the right track and actually reading or writing the data
 - Computes and attaches **checksums** to each sector to verify that data is read back correctly
 - If data is corrupted, with very high probability stored checksum won't match recomputed checksum
 - Ensures successful writing by reading back sector after writing it
 - Performs **remapping of bad sectors**



Performance Measures of Disks

- **Access time** – the time it takes from when a read or write request is issued to when data transfer begins. Consists of:
 - **Seek time** – time it takes to reposition the arm over the correct track.
 - Average seek time is 1/2 the worst case seek time.
 - Would be 1/3 if all tracks had the same number of sectors, and we ignore the time to start and stop arm movement
 - 4 to 10 milliseconds on typical disks
 - **Rotational latency** – time it takes for the sector to be accessed to appear under the head.
 - 4 to 11 milliseconds on typical disks (5400 to 15000 r.p.m.)
 - Average latency is 1/2 of the above latency.
 - Overall latency is 5 to 20 msec depending on disk model
- **Data-transfer rate** – the rate at which data can be retrieved from or stored to the disk.
 - 25 to 200 MB per second max rate, lower for inner tracks



Performance Measures (Cont.)

- **Disk block** is a logical unit for storage allocation and retrieval
 - 4 to 16 kilobytes typically
 - Smaller blocks: more transfers from disk
 - Larger blocks: more space wasted due to partially filled blocks
- **Sequential access pattern**
 - Successive requests are for successive disk blocks
 - Disk seek required only for first block
- **Random access pattern**
 - Successive requests are for blocks that can be anywhere on disk
 - Each access requires a seek
 - Transfer rates are low since a lot of time is wasted in seeks
- **I/O operations per second (IOPS)**
 - Number of random block reads that a disk can support per second
 - 50 to 200 IOPS on current generation magnetic disks



Performance Measures (Cont.)

- **Mean time to failure (MTTF)** – the average time the disk is expected to run continuously without any failure.
 - Typically 3 to 5 years
 - Probability of failure of new disks is quite low, corresponding to a “theoretical MTTF” of 500,000 to 1,200,000 hours for a new disk
 - E.g., an MTTF of 1,200,000 hours for a new disk means that given 1000 relatively new disks, on an average one will fail every 1200 hours
 - MTTF decreases as disk ages
- **Annualized Failure Rate (AFR):** $= (365 \times 24 / \text{MTTF}) \times 100\%$
 - $\text{MTTF} = 1,200,000 \rightarrow \text{AFR} = 0.73\%$
- Suppose MTTF is 1,200,000 hours for a disk. Then, in a system with 1000 disks, how often will a disk fail on average?
 - Answer: on average one will fail every 1200 hours (50 days)
 - Equivalently, 7.3 disks per year



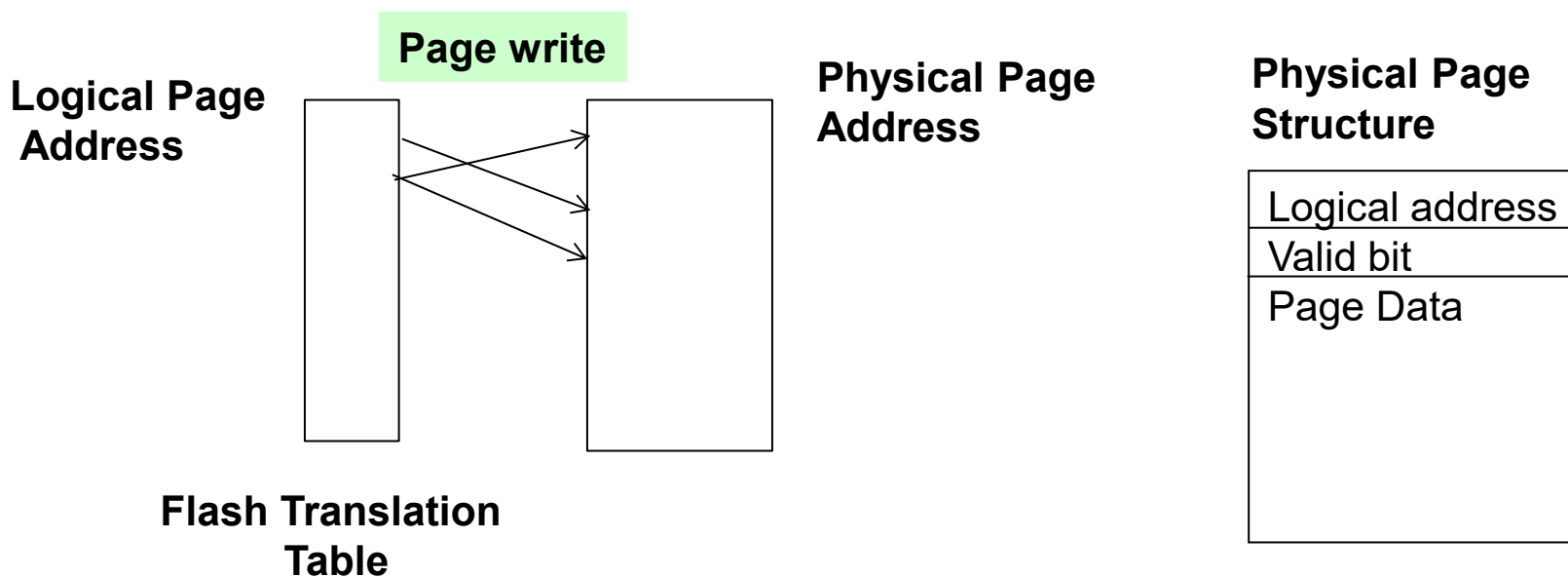
Flash Storage

- NOR flash vs NAND flash
- NAND flash
 - used widely for storage, cheaper than NOR flash
 - requires page-at-a-time read (page: 512 bytes to 4 KB)
 - 20 to 100 microseconds for a page read
 - Not much difference between sequential and random read
 - Page can only be written once
 - Must be erased to allow rewrite
- **Solid state disks**
 - Use standard block-oriented disk interfaces, but store data on multiple flash storage devices internally
 - Transfer rate of up to 500 MB/sec using SATA, and up to 3 GB/sec using NVMe PCIe



Flash Storage (Cont.)

- Erase happens in units of **erase block**
 - Takes 2 to 5 milliseconds
 - Erase block typically 256 KB to 1 MB (128 to 256 pages)
- **Remapping** of logical page addresses to physical page addresses avoids waiting for erase
- **Flash translation table** tracks mapping
 - also stored in a label field of flash page
 - remapping carried out by **flash translation layer**





Flash Storage (Cont.)

- SLC After about 1,00,000 erases (SLC Flash) to as low as 10,000 or 1000 erases (TLC/QLC Flash) erase block becomes unreliable and cannot be used
 - **wear leveling:** store infrequently updated (“cold”) data in blocks that have been erased many times already



Source: Kingston.com

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SSD Performance Metrics

- Random reads/writes per second
 - Typical 4 KB reads: 10,000 reads per second (10,000 IOPS)
 - Typical 4KB writes: 40,000 IOPS
 - SSDs support parallel reads
 - Typical 4KB reads:
 - 100,000 IOPS with 32 requests in parallel (QD-32) on SATA
 - 350,000 IOPS with QD-32 on NVMe PCIe
 - Typical 4KB writes:
 - 100,000 IOPS with QD-32, even higher on some models
- Data transfer rate for sequential reads/writes
 - 400 MB/sec for SATA3, 2 to 3 GB/sec using NVMe PCIe
- **Hybrid disks:** combine small amount of flash cache with larger magnetic disk



Storage Class Memory

- 3D-XPoint memory technology pioneered by Intel
- Available as Intel Optane
 - SSD interface shipped from 2017
 - Allows lower latency than flash SSDs
 - Non-volatile memory interface announced in 2018
 - Supports direct access to words, at speeds comparable to main-memory speeds



RAID

- **RAID: Redundant Arrays of Independent Disks**
 - disk organization techniques that manage a large numbers of disks, providing a view of a single disk of
 - **high capacity** and **high speed** by using multiple disks in parallel,
 - **high reliability** by storing data redundantly, so that data can be recovered even if a disk fails
- The chance that some disk out of a set of N disks will fail is much higher than the chance that a specific single disk will fail.
 - E.g., a system with 100 disks, each with MTTF of 100,000 hours (approx. 11 years), will have a system MTTF of 1000 hours (approx. 41 days)
 - Techniques for using redundancy to avoid data loss are critical with large numbers of disks



Improvement of Reliability via Redundancy

- **Redundancy** – store extra information that can be used to rebuild information lost in a disk failure
- E.g., **Mirroring** (or **shadowing**)
 - Duplicate every disk. Logical disk consists of two physical disks.
 - Every write is carried out on both disks
 - Reads can take place from either disk
- **Mean time to data loss** depends on mean time to failure, and **mean time to repair**
 - E.g. MTTF of 100,000 hours, mean time to repair of 10 hours gives mean time to data loss of 500×10^6 hours (or 57,000 years) for a mirrored pair of disks (ignoring dependent failure modes)



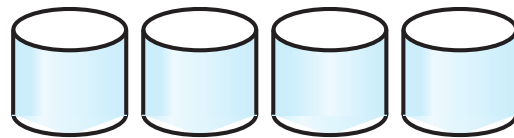
Improvement in Performance via Parallelism

- Goals of parallelism in a disk system:
 1. Load balance multiple small accesses to increase throughput
 2. Parallelize large accesses to reduce response time.
 3. Improve transfer rate by striping data across multiple disks.
- **Bit-level striping**
 - Not used in practice
- **Block-level striping** – with n disks, block i of a file goes to disk $(i \bmod n) + 1$
 - Requests for different blocks can run in parallel if the blocks reside on different disks
 - A request for a long sequence of blocks can utilize all disks in parallel

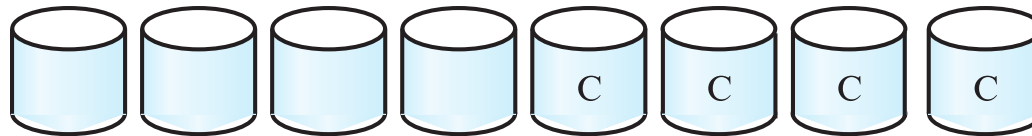


RAID Levels

- **RAID Level 0:** Block striping; non-redundant.
 - Used in high-performance applications where data loss is not critical.
- **RAID Level 1:** Mirrored disks with block striping
 - Offers best write performance.
 - Popular for applications such as storing log files in a database system.



(a) RAID 0: nonredundant striping



(b) RAID 1: mirrored disks



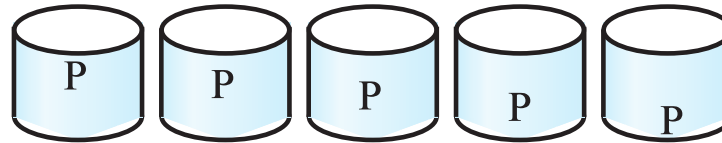
RAID Levels (Cont.)

- **Parity blocks:** Parity block j stores XOR of bits from block j of each disk
 - When writing data to a block j , parity block j must also be computed and written to disk
 - Can be done by using old parity block, old value of current block and new value of current block (2 block reads + 2 block writes)
 - Or by recomputing the parity value using the new values of blocks corresponding to the parity block
 - More efficient for writing large amounts of data sequentially
 - To recover data for a block, compute XOR of bits from all other blocks in the set including the parity block



RAID Levels (Cont.)

- **RAID Level 5: Block-Interleaved Distributed Parity;** partitions data and parity among all $N + 1$ disks, rather than storing data in N disks and parity in 1 disk.
 - E.g., with 5 disks, parity block for n th set of blocks is stored on disk $(n \bmod 5) + 1$, with the data blocks stored on the other 4 disks.



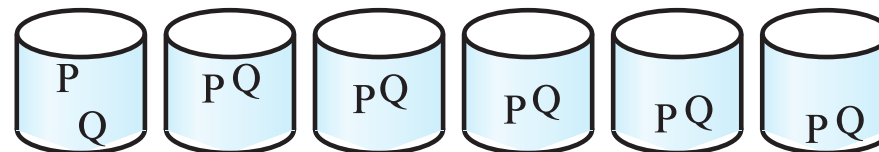
(c) RAID 5: block-interleaved distributed parity

P0	0	1	2	3
4	P1	5	6	7
8	9	P2	10	11
12	13	14	P3	15
16	17	18	19	P4



RAID Levels (Cont.)

- **RAID Level 5 (Cont.)**
 - Block writes occur in parallel if the blocks and their parity blocks are on different disks.
- **RAID Level 6: P+Q Redundancy** scheme; similar to Level 5, but stores two error correction blocks (P, Q) instead of single parity block to guard against multiple disk failures.
 - Better reliability than Level 5 at a higher cost
 - Becoming more important as storage sizes increase



(d) RAID 6: P + Q redundancy



RAID Levels (Cont.)

- **Other levels (not used in practice):**
 - **RAID Level 2:** Memory-Style Error-Correcting-Codes (ECC) with bit striping.
 - **RAID Level 3:** Bit-Interleaved Parity
 - **RAID Level 4:** Block-Interleaved Parity; uses block-level striping, and keeps a parity block on a separate ***parity disk*** for corresponding blocks from N other disks.
 - RAID 5 is better than RAID 4, since with RAID 4 with random writes, parity disk gets much higher write load than other disks and becomes a bottleneck



Choice of RAID Level

- Factors in choosing RAID level
 - Monetary cost
 - Performance: Number of I/O operations per second, and bandwidth during normal operation
 - Performance during failure
 - Performance during rebuild of failed disk
 - Including time taken to rebuild failed disk
- RAID 0 is used only when data safety is not important
 - E.g. data can be recovered quickly from other sources



Choice of RAID Level (Cont.)

- Level 1 provides much better write performance than level 5
 - Level 5 requires at least 2 block reads and 2 block writes to write a single block, whereas Level 1 only requires 2 block writes
- Level 1 had higher storage cost than level 5
- Level 5 is preferred for applications where writes are sequential and large (many blocks), and need large amounts of data storage
- RAID 1 is preferred for applications with many random/small updates
- Level 6 gives better data protection than RAID 5 since it can tolerate two disk (or disk block) failures
 - Increasing in importance since latent block failures on one disk, coupled with a failure of another disk can result in data loss with RAID 1 and RAID 5.



Hardware Issues

- **Software RAID:** RAID implementations done entirely in software, with no special hardware support
- **Hardware RAID:** RAID implementations with special hardware
 - Use non-volatile RAM to record writes that are being executed
 - Beware: power failure during write can result in corrupted disk
 - E.g. failure after writing one block but before writing the second in a mirrored system
 - Such corrupted data must be detected when power is restored
 - Full scan of disk may be required!
 - NV-RAM helps to efficiently detected potentially corrupted blocks



Hardware Issues (Cont.)

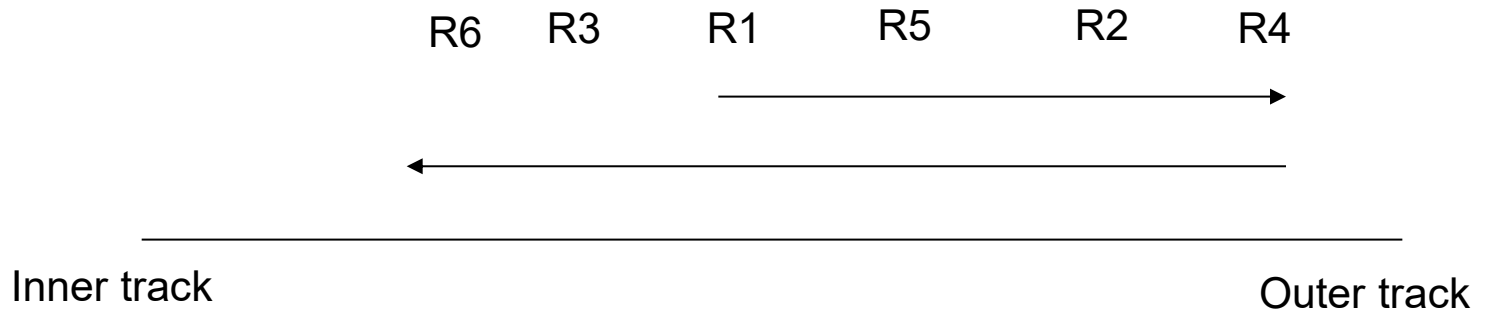
- **Latent sector failures:** data successfully written earlier gets damaged
 - can result in data loss even if only one disk fails
- **Data scrubbing:**
 - continually scan for latent failures, and recover from copy/parity
- **Hot swapping:** replacement of disk while system is running, without power down
 - Supported by some hardware RAID systems,
 - reduces time to recovery, and improves availability greatly
- **Spare disks** are kept online, and used as replacements for failed disks immediately on detection of failure
 - Reduces time to recovery greatly
- To avoid single point of failure
 - Redundant power supplies with UPS backup
 - Multiple network controllers/network interconnections





Optimization of Disk-Block Access

- **Buffering:** in-memory buffer to cache disk blocks
- **Read-ahead:** Read extra blocks from a track in anticipation that they will be requested soon
- **Disk-arm-scheduling** algorithms re-order block requests so that disk arm movement is minimized
 - **elevator algorithm**





End of Chapter 12

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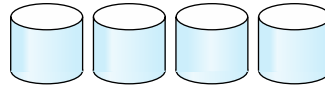


Magnetic Tapes

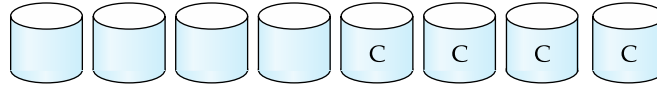
- Hold large volumes of data and provide high transfer rates
 - Few GB for DAT (Digital Audio Tape) format, 10-40 GB with DLT (Digital Linear Tape) format, 100 GB+ with Ultrium format, and 330 GB with Ampex helical scan format
 - Transfer rates from few to 10s of MB/s
- Tapes are cheap, but cost of drives is very high
- Very slow access time in comparison to magnetic and optical disks
 - limited to sequential access.
 - Some formats (Accelis) provide faster seek (10s of seconds) at cost of lower capacity
- Used mainly for backup, for storage of infrequently used information, and as an off-line medium for transferring information from one system to another.
- Tape jukeboxes used for very large capacity storage
 - Multiple petabytes (10^{15} bytes)



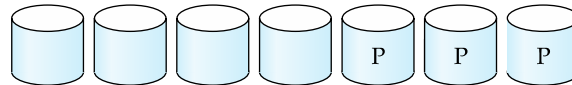
Figure 10.03



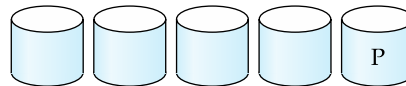
(a) RAID 0: nonredundant striping



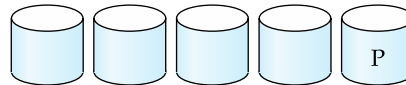
(b) RAID 1: mirrored disks



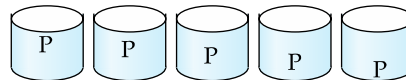
(c) RAID 2: memory-style error-correcting codes



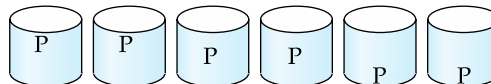
(d) RAID 3: bit-interleaved parity



(e) RAID 4: block-interleaved parity



(f) RAID 5: block-interleaved distributed parity



(g) RAID 6: P + Q redundancy



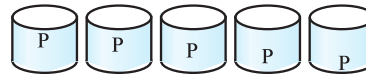
Disk 1	Disk 2	Disk 3	Disk 4
B_1	B_2	B_3	B_4
P_1	B_5	B_6	B_7
B_8	P_2	B_9	B_{10}
$\vdots$	$\vdots$	$\vdots$	$\vdots$



(a) RAID 0: nonredundant striping



(b) RAID 1: mirrored disks



(c) RAID 5: block-interleaved distributed parity



(d) RAID 6: P + Q redundancy